

SD Seminar 1

27 Feb 2025

1. Clase de complexități

ex 1

complexități
for

ex 2

$$f(n) \in \Theta(g(n)) \quad \Leftrightarrow \quad \begin{cases} f(n) \in O(g(n)) \\ f(n) \in \Omega(g(n)) \end{cases}$$

ex 3

$$\max(f(n), g(n)) \in \Theta(f(n) + g(n))$$

ex 4

$$2^{n+1} \in O(2^n)$$

Ⓐ

$$2^{2^n} \in O(2^n)$$

Ⓣ

ex 5

a, b constante naturale ≥ 1
+

$$(n+a)^b \in O(n^b)$$

#

ex 6

$$\left. \begin{array}{l} f(n) \in O(g(n)) \\ g(n) \in O(f(n)) \end{array} \right\} \Rightarrow h(n) \in \Omega(f(n))$$

#

ex 7

confirmati non informati

$$a) f(n) \in O(g(n)) \Rightarrow g(n) \in O(f(n))$$

$$b) f(n) \in O(g(n)) \Rightarrow 2^{f(n)} \in O(2^{g(n)})$$

$$c) o(g(n)) \cap w(g(n)) = \emptyset$$

2. Recurenta

ex 1

$$T(n) = T(n-1) + n$$

#

ex 2

$$a) T(n) = 2 \cdot T\left(\frac{n}{2}\right) + 1$$

$$b) T(n) = 2 \cdot T\left(\frac{n}{2}\right) + \sqrt{n}$$

$$c) T(n) = 2 \cdot T\left(\frac{n}{2}\right) + n$$

$$d) T(n) = 2 \cdot T\left(\frac{n}{2}\right) + n^2$$

ex 3

Cantorea binară

#

ex 4

$$T(n) = 4 T\left(\frac{n}{3}\right) + n$$

#

ex 5

$$T(n) = 3 \cdot T\left(\frac{n}{2}\right) + n$$

#

Clase de complexitate

$$O(g(n)) = \{ f(n) \mid \exists c > 0, n_0 > 0 \text{ a.i.} \}$$

$$0 \leq f(n) \leq c \cdot g(n), \forall n \geq n_0 \}$$

$$\Omega(g(n)) = \{ f(n) \mid \exists c > 0, n_0 > 0 \text{ a.i.} \}$$

lower bound

$$0 \leq c \cdot g(n) \leq f(n) \quad \forall n \geq n_0 \}$$

$$n^2 \in O(n^3) \quad (A)$$

$$n^2 \in O(n) \quad (\text{f})$$

$$n^2 \in \Omega(n) \quad (A)$$

$$n \in \Omega(n^2) \quad (\text{f})$$

$$\Theta(g(n)) = \{ f(n) \mid \exists c_1, c_2 > 0 \text{ a.i.} \}$$

$$c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n), \forall n \geq n_0 \}$$

$$o(g(n)) = \{ f(n) \mid \underline{\forall c > 0} \quad \exists n_c > 0 \text{ a.i.} \}$$

$$f(n) < c \cdot g(n) \quad \forall n \geq n_0$$

$$\omega(g(n)) = \{ f(n) \mid \underline{\forall c > 0} \quad \exists n_c > 0 \text{ a.i.} \}$$

$$f(n) > c \cdot g(n) \quad \forall n \geq n_0$$

ex 1

complexitate
for

$$\text{for (int } i = 1; i \leq n; i++) \rightarrow O(n)$$

$$i = i * 2; \rightarrow O(\log n)$$

$$i * i \leq n; i++ \rightarrow O(\sqrt{n})$$



n 2

$$f(n) \in \Theta(g(n)) \quad \Leftrightarrow \quad \begin{cases} f(n) \in O(g(n)) \\ f(n) \in \Omega(g(n)) \end{cases}$$

Id:

" $\Rightarrow$ "

$$f(n) \in \Theta(g(n))$$

+

$$\Rightarrow \exists c_1, c_2 > 0 \quad \text{a.i.} \quad c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n) \quad \forall n \geq n_0$$

$$\Rightarrow \begin{cases} c_1 \cdot g(n) \leq f(n) & \Rightarrow f(n) \in \Omega(g(n)) \\ f(n) \leq c_2 \cdot g(n) & \Rightarrow f(n) \in O(g(n)) \end{cases}$$

" $\Leftarrow$ "

$$f(n) \in O(g(n)) \quad (1)$$

$$f(n) \in \Omega(g(n)) \quad (2)$$

$$(1) \Rightarrow \exists c_1 > 0 \quad \text{a.i.} \quad f(n) \leq c_1 \cdot g(n) \quad \forall n \geq n_1$$

$$(2) \Rightarrow \exists c_2 > 0 \quad \text{a.i.} \quad c_2 \cdot g(n) \leq f(n) \quad \forall n \geq n_2$$

$$\Rightarrow c_2 \cdot g(n) \leq f(n) \leq c_1 \cdot g(n) \quad , \quad \forall n > \max(n_1, n_2)$$

$$\Rightarrow f(n) \in \Theta(g(n))$$



ex 3

$$\max(f(n), g(n)) \in \theta(f(n) + g(n))$$

Sol:

I upper bound

$$\max(f(n), g(n)) \leq f(n) + g(n)$$

$$\Rightarrow \underline{\max(f(n), g(n)) \in O(f(n) + g(n))}$$

II lower bound

$$f(n) \leq \max(f(n), g(n))$$

$$g(n) \leq \max(f(n), g(n))$$

$$f(n) + g(n) \leq 2 \cdot \max(f(n), g(n)) \quad | : 2$$

$$c = \frac{1}{2} (f(n) + g(n)) \leq \max(f(n), g(n))$$

$$\Rightarrow \underline{\max(f(n), g(n)) \in \Omega(f(n) + g(n))}$$

I ' II $\Rightarrow$

$$\max(f(n), g(n)) \in \theta(f(n) + g(n))$$

□

ex 4

$$2^{n+1} \in O(2^n)$$

Ⓐ

$$2^{2^n} \in O(2^n)$$

Ⓕ

ex 5

a, b constante naturale ≥ 1

+

$$(n+a)^b \in O(n^b)$$

#

Sol:

Binomial lui Newton

$$(n+a)^b = C_b^0 n^b a^0 + C_b^1 n^{b-1} a + \dots + C_b^b a^b$$

I lower bound

$$(n+a)^b \geq C_b^0 n^b a^0 = n^b$$

$$\Rightarrow (n+a)^b \in \Omega(n^b)$$

II upper bound

$$(n+a)^b \leq C_b^0 n^b a^0 + C_b^1 n^{b-1} a^1 + \dots + C_b^b n^0 a^b$$

$$\leq n^b (C_b^0 a^0 + C_b^1 a^1 + \dots + C_b^b a^b)$$

$$\leq n^b \cdot (1+a)^b$$

↗ constantă

$$\Rightarrow (n+a)^b \in O(n^b)$$

$$\Rightarrow (n+a)^b \in \Theta(n^b)$$

ex 6

$$\left. \begin{array}{l} f(n) \in O(g(n)) \\ g(n) \in O(h(n)) \end{array} \right\} \Rightarrow \begin{array}{l} h(n) \in \Omega(f(n)) \\ \neq \end{array}$$

Sol:

$$f(n) \in O(g(n))$$

$$\Rightarrow \exists \begin{array}{l} c_1 > 0 \\ n_1 > 0 \end{array} \quad \text{a.i.} \quad f(n) \leq c_1 \cdot g(n) \quad \forall n \geq n_1$$

$$g(n) \in O(h(n))$$

$$\Rightarrow \exists \begin{array}{l} c_2 > 0 \\ n_2 > 0 \end{array} \quad \text{a.i.} \quad g(n) \leq c_2 \cdot h(n) \quad \forall n \geq n_2$$

$$\Rightarrow f(n) \leq c_1 \cdot c_2 \cdot h(n) \quad \forall n \geq \max(n_1, n_2)$$

$$\Leftrightarrow h(n) \geq \frac{1}{c_1 \cdot c_2} \cdot f(n) \quad \forall n \geq \max(n_1, n_2)$$

$$\Rightarrow h(n) \in \Omega(f(n))$$



ex 7

confirmati non informati

$$a) f(n) \in O(g(n)) \Rightarrow g(n) \in O(f(n))$$

$$b) f(n) \in O(g(n)) \Rightarrow 2^{f(n)} \in O(2^{g(n)})$$

$$c) o(g(n)) \cap w(g(n)) = \emptyset$$

Sol:

$$a) f(n) \in O(g(n)) \Rightarrow g(n) \in O(f(n))$$

Contro example

$$f(n) = n$$

$$n \in O(n^2)$$

$$g(n) = n^2$$

$$n^2 \notin O(n)$$

$\Rightarrow$ FALS

$$b) f(n) \in O(g(n))$$

$$\Rightarrow 2^{f(n)} \in O(2^{g(n)})$$

Contro example

$$f(n) = 2n$$

$$2n \in O(n)$$

$$g(n) = n$$

$$2^{2n} \notin O(2^n)$$

$\Rightarrow$ FALS

$$c) \quad o(g(n)) \cap w(g(n)) = \emptyset$$

Reducere la absurd

$$p_n. \exists f(n) \in o(g(n)) \cap w(g(n))$$

$$f(n) \in o(g(n))$$

$$\Rightarrow \forall c_1 > 0 \quad \exists n_1 > 0 \quad \text{a.i.} \quad f(n) < c_1 \cdot g(n) \quad \forall n \geq n_1$$

$$f(n) \in w(g(n))$$

$$\Rightarrow \forall c_2 > 0 \quad \exists n_2 > 0 \quad \text{a.i.} \quad c_2 \cdot g(n) < f(n) \quad \forall n \geq n_2$$

$$\Rightarrow \exists c > 0 \quad \text{a.i.} \quad c \cdot g(n) < f(n) < c \cdot g(n)$$

n_1, n_2

$$\forall n \geq \max(n_1, n_2)$$

Contradictie

$$\Rightarrow \text{ADEVARAT}$$



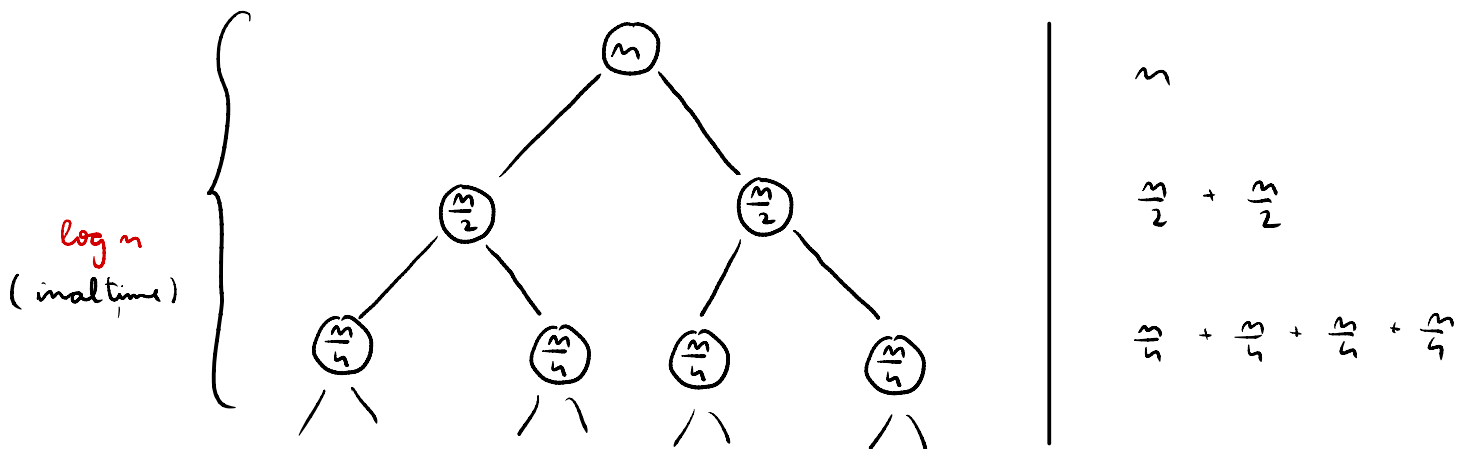
Recurențe

$T(n)$ = complexitatea algoritmului pt un input de dimensiune n

Merge Sort

$$T(n) = 2 T\left(\frac{n}{2}\right) + O(n) \in O(n \log n)$$

I Arborele de recurență



II Teorema Master

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + O(n^d), \quad a, b, d \text{ et}$$

$$T(n) = \begin{cases} \Theta(n^d), & a < b^d \\ \Theta(n^d \log n), & a = b^d \\ \Theta(n^{\log_b a}), & a > b^d \end{cases}$$

III Metoda Substitției

$$T(n) \leq \underline{c \cdot n \log n}$$

$$T(1), T(2), \dots, T(n-1) \Rightarrow T(n)$$

$$T\left(\frac{n}{2}\right) = c \cdot \frac{n}{2} \log_2 \frac{n}{2}$$

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + n$$

$$= \cancel{2} \cdot c \cdot \cancel{\frac{n}{2}} \log_2 \frac{n}{2} + n$$

$$= c \cdot n (\log_2 n - \log_2 2) + n$$

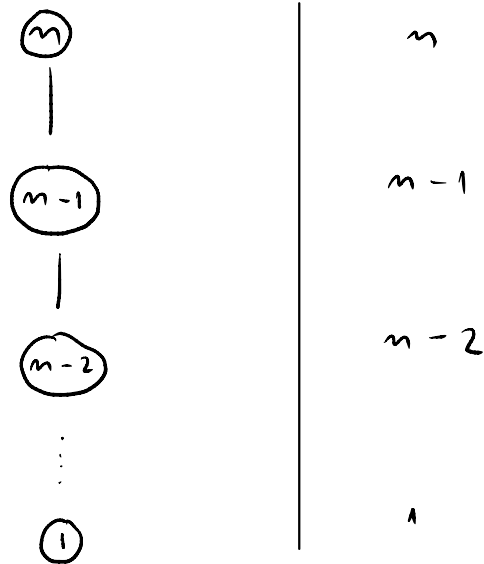
$$= c \cdot n \cdot \log_2 n - c \cdot n + n$$

$$= c \cdot n \cdot \log_2 n - n(c - 1) \leq c \cdot n \log n$$

$$\forall c \geq 1$$

ex 1

$$T(n) = T(n-1) + n$$



$$n + n-1 + n-2 + \dots + 1 = \frac{n(n+1)}{2} \approx n^2$$

$$T(n) \leq c \cdot n^2$$

$$\begin{array}{ccc} T(1), T(2), \dots, T(n-1) & \Rightarrow & T(n) \\ + & & + \end{array}$$

$$T(n-1) \leq c \cdot (n-1)^2 = c(n^2 - 2n + 1)$$

$$\begin{aligned} T(n) &= c \cdot (n^2 - 2n + 1) + n \\ &= c \cdot n^2 - c \cdot 2n + c + n \\ &= c \cdot n^2 + n(1 - 2c) + c \leq c \cdot n^2 \quad \forall n \geq \frac{1}{2} \end{aligned}$$

ex 2

a) $T(n) = 2 \cdot T\left(\frac{n}{4}\right) + 1$

b) $T(n) = 2 \cdot T\left(\frac{n}{4}\right) + \sqrt{n}$

c) $T(n) = 2 \cdot T\left(\frac{n}{4}\right) + n$

d) $T(n) = 2 \cdot T\left(\frac{n}{4}\right) + n^2$

Sol:

a) $T(n) = 2 \cdot T\left(\frac{n}{4}\right) + 1$

$a = 2$

$a \quad \square \quad b^a$

$b = 4$

$2 > 4^0$

$d = 1$

$\Rightarrow \log_2 3 \Rightarrow \Theta(n^{\log_2 3})$

$T(n) \in O(n^{\log_4 2}) = O(n^{\frac{1}{2}})$

$T(n) \in O(\sqrt{n})$

b) $T(n) = 2 \cdot T\left(\frac{n}{4}\right) + \sqrt{n}$

$a = 2$

$a \quad \square \quad b^a \rightarrow \log_2 2$

$b = 4$

$2 = 4^{\frac{1}{2}}$

$d = \frac{1}{2}$

$2 = \sqrt{4} = 2$

$T(n) \in O(\sqrt{n} \log n)$

$$c) \quad T(n) = 2 \cdot T\left(\frac{n}{4}\right) + n$$

$$a = 2$$

$$b = 4$$

$$d = 1$$

$$a \leq b^d \Rightarrow \log 1$$

$$2 \leq 4^1$$

$$T(n) \in \Theta(n)$$

$$d) \quad T(n) = 2 \cdot T\left(\frac{n}{4}\right) + n^2$$

$$a = 2$$

$$b = 4$$

$$d = 2$$

$$a \leq b^d \Rightarrow \log 1$$

$$2 \leq 4^2$$

$$T(n) \in O(n^2)$$



ex 3

cantarea binară
#

$$T(n) = T\left(\frac{n}{2}\right) + 1$$

$$\log_2 n \left\{ \begin{array}{c|c} 3 & 1 \\ 2 & 1 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{array} \right.$$

$$T(n) \leq c \cdot \log n$$

#

P_n. $T(1), T(2), \dots, T(n-1)$ adur

Dem $T(n)$

$$T\left(\frac{n}{2}\right) \leq c \cdot \log_2 \frac{n}{2}$$

$$\begin{aligned} T(n) &= T\left(\frac{n}{2}\right) + 1 \leq c \cdot \log_2 \frac{n}{2} + 1 = \\ &= c \cdot (\log_2 n - \log_2 2) + 1 \\ &= c \cdot \log_2 n - c + 1 \\ &= c \cdot \log_2 n - (c - 1) \leq c \cdot \log_2 n \\ &\quad \forall c > 1 \end{aligned}$$

ex 4

$$T(n) = 4 T\left(\frac{n}{3}\right) + n$$

#

$$a = 4$$

$$b = 3$$

$$d = 1$$

$$a > b^d \Rightarrow \log_3 4$$

$$4 > 3^1$$

$$T(n) \in \Theta(n^{\log_3 4})$$

$$T(n) \leq c \cdot n^{\log_3 4}$$

#

$$T\left(\frac{n}{3}\right) \leq c \cdot \left(\frac{n}{3}\right)^{\log_3 4}$$

$$T(n) = h \cdot T\left(\frac{n}{3}\right) + n \leq h \cdot c \cdot \left(\frac{n}{3}\right)^{\log_3 4} + n =$$

$$= h \cdot c \cdot \frac{n^{\log_3 4}}{3^{\log_3 4}} + n$$

$$= \cancel{h} \cdot c \cdot \frac{n^{\log_3 4}}{\cancel{4}} + n$$

$$= c \cdot n^{\log_3 4} + n$$

← ne avonjenjā
⇒ refaer!!

$$T(n) \leq c \cdot n^{\log_3 4} - \underline{d \cdot n}$$

$$T\left(\frac{n}{3}\right) \leq c \cdot \left(\frac{n}{3}\right)^{\log_3 4} - d \cdot \frac{n}{3}$$

$$T(n) = h \cdot \left(c \cdot \left(\frac{n}{3}\right)^{\log_3 4} - d \cdot \frac{n}{3} \right) + n$$

$$= \cancel{h} \cdot c \cdot \frac{n^{\log_3 4}}{\cancel{4}} - h d \frac{n}{3} + n$$

$$= c \cdot n^{\log_3 4} - d n - \frac{1}{3} d n + n$$

$$= c \cdot n^{\log_3 4} - d n + n \left(1 - \frac{1}{3} d\right) \leq c \cdot n^{\log_3 4} - d n$$

$$\forall d \geq 3$$

$$\forall c \geq 0$$

$$c \cdot n^{\log_3 4} - n \left(\underline{\frac{4}{3} d - 1} \right) \leq c \cdot n^{\log_3 4} - \underline{n d}$$

$$d \leq \frac{4}{3} d - 1$$

$$3d \leq 4d - 3$$

$$3 \leq d$$



ex 5

Fie A, B două nr. în baza 10
cu n cifre

Algoritmul clasic de înmulțire ale celor nr.
este de $O(n^2)$

Algoritmul lui Karatsuba $\Rightarrow O(n^{\log_2 3})$

Fast Fourier Transform (FFT) $\Rightarrow O(n \log n)$

Din teorema împărțirii cu rest avem

$$A = x_A \cdot 10^{\frac{n}{2}} + y_A$$

$$B = x_B \cdot 10^{\frac{n}{2}} + y_B$$

$x_A, y_A, x_B, y_B \rightarrow$ cel mult $\frac{n}{2}$ cifre

$$A \cdot B = (x_A \cdot 10^{\frac{n}{2}} + y_A) \cdot (x_B \cdot 10^{\frac{n}{2}} + y_B)$$

$$= x_A x_B \cdot 10^n + x_A y_B \cdot 10^{\frac{n}{2}} + y_A x_B \cdot 10^{\frac{n}{2}} + y_A y_B$$

$$= \underline{x_A x_B} \cdot 10^n + 10^{\frac{n}{2}} \cdot (\underline{x_A y_B} + \underline{y_A x_B}) + \underline{y_A y_B}$$

$$T(n) = 4 \cdot T\left(\frac{n}{2}\right) + n \Rightarrow O(n^2)$$

$$\underline{x_A x_B} \cdot 10^n + 10^{\frac{n}{2}} \cdot (x_A y_B + y_A x_B) + \underline{y_A y_B} \quad \sim \text{deja calculate}$$

$$x_A y_B + y_A x_B = (x_A + y_A)(x_B + y_B) - x_A x_B - y_A y_B$$

$\Rightarrow$ din 4 înmulțiri am făcut 3

$$\Rightarrow T(n) = 3 \cdot T\left(\frac{n}{2}\right) + n$$

##

$$a = 3$$

$$b = 2$$

$$d = 1$$

$$a \quad \boxed{>} \quad b^d \quad \log_2 3$$

$$3 \quad \boxed{>} \quad 2^1$$

$$T(n) \in \Theta(n^{\log_2 3})$$

$$T(n) \in O(n^{\log_2 3})$$